DIGITAL LIBRARY SEARCH SYSTEM

Data Structures and Algorithms - 3 (25CS2103E)

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Team Number	____________________
Team Members	_____________________________________
Guide	Dr. V. Sireesha

ABSTRACT

The **Digital Library Search System** is developed to simplify the process of searching and retrieving digital library resources from a centralized repository. Libraries contain a vast collection of books, journals, magazines, and research papers, making manual searching time-consuming and inefficient. The proposed system enables users to search resources using multiple parameters such as **title, author, ISBN, category, and keywords**, ensuring quick and accurate retrieval of information. The project focuses on implementing efficient searching techniques and appropriate data structures to organize and access library records effectively. The initial version emphasizes basic search functionality while providing a scalable foundation for future enhancements, including intelligent search features similar to those used in modern digital libraries. By improving search efficiency, reducing retrieval time, and enhancing accessibility, the proposed system offers an effective solution for managing and locating digital library resources, demonstrating the practical application of Data Structures and Algorithms in information retrieval systems.

PROBLEM STATEMENT

Libraries contain thousands of books, journals, magazines, and research papers. Finding the required resource quickly becomes difficult when an efficient search system is not available.

Key Challenges

- Large volume of digital library resources.
- Manual or basic searching can be slow and time-consuming.
- Users need different ways to locate a resource.
- Search results should be retrieved quickly and accurately.
- The system should be capable of supporting improved search features in the future.

Proposed Solution: Build a centralized digital library repository where users can search resources by title, author, ISBN, category, or keywords using efficient searching techniques.

PROPOSED FUNCTIONALITIES

Digital Repository	Store and organize books, journals, magazines, and research papers.
Multi-Parameter Search	Search using title, author, ISBN, category, and keywords.
Efficient Retrieval	Return matching records quickly using suitable data structures and searching techniques.
Resource Organization	Maintain structured and accessible library records.
User-Friendly Search	Provide a simple interface for locating required resources.
Future Intelligent Search	Extend the system later with improved search relevance and intelligent search features.

TOOLS AND TECHNOLOGIES TO BE USED

Programming Language	Java
Data Structures / DSA	Arrays, linked structures, hashing concepts, searching and sorting techniques; advanced string-search techniques can be added as the project evolves.
Database	MySQL
Database Connectivity	JDBC
Development Environment	VS Code / IntelliJ IDEA / Eclipse
Database Tool	MySQL Workbench
Version Control	Git and GitHub

EXPECTED OUTCOME

The completed Digital Library Search System is expected to provide an organized and efficient way to store, search, and retrieve digital library resources.

- Reduce the time required to locate library resources.
- Enable search through title, author, ISBN, category, and keywords.
- Improve the speed and accuracy of resource retrieval.
- Maintain an organized centralized repository of digital resources.
- Demonstrate the practical use of Data Structures and Algorithms in an information retrieval application.
- Provide a foundation for future intelligent search features similar to modern digital libraries.

Final Goal: A scalable, efficient, and user-friendly digital library search solution.